

Alex Kurkar

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EDUCATION

University of Nottingham

BSc Computer Science: Predicted 1:1, 90% average

Nottingham, UK

Sept 2025 – Jun 2028

Altrincham Grammar School for Boys

A-Levels: Mathematics A*, Physics A, Further Maths A, Computer Science A

Manchester, UK

Sept 2023 – Jun 2025

EXPERIENCE

Code Computerlove

Product Development Work Experience

Manchester, UK

Aug 2024

- Prototyped a consumer-facing application in a cross-functional team, contributing to user-centred UI/UX in Figma and full-stack implementation.
- Worked within agile sprints (planning, standups, retrospectives), iterating on features using user feedback and product requirements.

Cheshire Datasystems Limited

Software Engineering Work Experience

Cheshire, UK

Aug 2023

- Shadowed hardware and software teams, gaining exposure to large-scale system architecture, implementation, and QA workflows.
- Observed the full software development lifecycle from requirements gathering through testing and deployment, strengthening understanding of reliable delivery practices.

COMPETITIONS & HACKATHONS

CHAINIQ | Procurement agent, START Hack 2026, St Gallen, Switzerland

Next.js, TypeScript, React, Tailwind, MySQL, Neo4j, OpenAI

Mar 2026

- Built an audit-ready procurement agent that turns messy free-text purchase requests into structured, policy-compliant supplier recommendations with transparent reasoning and escalation logic.

ENS Pay | Crypto payments MVP, BSA Stablecoins & Payments, EPFL, Switzerland

Python, aiogram, aiohttp, Telegram Mini App, TON Connect, HTML/JS

Mar 2026

- Built a Telegram-based crypto payments MVP resolving ENS names to wallet-ready TON payment flows, with a Python bot and Mini App handling wallet connection, links, and QR checkout.

Use-Change Hunter | Real-estate opportunity analysis, HackLondon 2026

Next.js, TypeScript, FastAPI, PostgreSQL/PostGIS, Redis, Docker, Leaflet

Feb 2026

- Prototyped a platform that scores development potential, risk, and confidence for properties by analysing nearby planning precedents and exporting scenario reports.

UKIEPC 2025 | ICPC UK & Ireland Programming Contest, Team LockedIn

Oct 2025

- Represented the University of Nottingham in the ICPC UK & Ireland Programming Contest; placed 142nd overall as part of team *LockedIn*.

Rubber Duck Agent | AI debugging companion, DurHack X & MLH AI Roadshow

Python, FastAPI, Local LLMs

2025

- Built an AI “rubber duck” debugging companion where a local LLM tracks how “locked in” a developer is from git and interaction signals; demoed live at the MLH AI Roadshow.

Paladins of Pi | Offline LLM dungeon master, HackNotts

Python, Ollama, Raspberry Pi

2025

- Led a team to build an offline LLM dungeon master on Raspberry Pi 5, delivering a web-based text adventure fully generated and served on-device.

PERSONAL PROJECTS

Pascal Interpreter & Bytecode VM | *Python, C, Bytecode*

Sept 2025 – Present

- Built a Pascal-like language toolchain from scratch with zero dependencies: a recursive-descent parser producing a full AST, a semantic analyser enforcing scoped symbol tables and static type checking, a tree-walking interpreter, and a stack-based bytecode compiler targeting a custom VM in C.
- VM executes emitted `.pbc` bytecode files with call frames, labels, jumps, and scoped variables; an interactive REPL supports running, compiling, and executing programs via CLI.

Neural Network from Scratch | *Python, NumPy*

Sept 2023 – Dec 2024

- Built a fully-connected neural network for digit classification using matrix-based backpropagation and SGD, achieving over 95% accuracy without high-level frameworks.

TECHNICAL SKILLS

Languages: Python, C, C#, Java, JavaScript, HTML/CSS

Frameworks & Libraries: React, Node.js, Flask, FastAPI, Next.js, aiogram, TensorFlow, PyTorch, NumPy, pandas, scikit-learn

Tools: Git, Docker, VS Code, IntelliJ, Jupyter Notebooks

Areas: ML systems, deep learning, time series, compilers/interpreters, competitive programming, crypto payments, hackathon prototyping